V-25—Initial inspection of electronic registers which incorporate multipoint calibration features, in liquid metering systems

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1.0 Purpose

This bulletin is intended to outline the Measurement Canada policy regarding the use and initial inspection of electronic registers which employ a means of adjustment that automatically selects a predetermined correction factor that corresponds to the rate of flow (multipoint calibration).

2.0 Scope

This bulletin is applicable to all initial inspections of metering systems which incorporate electronic registers that have the option of multipoint calibration. It does not apply to retail motor fuel dispensers, mass flow meters (Coriolis meters) or Magnetic Flow Meters. As well, it does not apply to subsequent inspections of these meters.

3.0 Background

Measurement Canada recognizes multipoint calibration as a viable method to improve the accuracy of metering systems. The use of such features has been allowed, with the understanding that it was only to be used to improve the performance of meters which would already meet the legislative requirements. There is reason to believe that the feature is also being used to enhance the performance of meters which would otherwise not meet those requirements. The area of concern lies with meters ability to measure accurately throughout their approved flowrate range. The difference between the accuracy at high flow rates and at slow flow rates is commonly called the spread. All meters currently approved in Canada were evaluated without the aid of multipoint calibration. When production meters which do not meet all requirements are presented for inspection, it is assumed that something has changed. The specific cause of meters not meeting the requirements throughout their rated flow rates is not clear. This may be as a result of defective parts in the new meters, a change in the design since the meters have been approved, or a change due to reduced drag on the meter due to the advent of electronic registers. In any case, the intention of allowing multipoint calibration was to improve the performance of meters which would already meet the legal requirements.

It was not the intent to allow multipoint calibration to be used to mask the performance of substandard meters in a way that makes them appear to meet the applicable requirements.

4.0 Policy

Measurement Canada or any party conducting inspections will determine if the meter in the system is capable of meeting the performance requirements, without the use of multipoint calibration feature, throughout its approved flow range. In addition it will be assured that means of adjustment are not being used to compensate for faulty installation or for worn or defective parts.

4.1 Additional tests

When a metering system which has the option of using multipoint calibration is presented for initial inspection, a fast and a slow test must be conducted using a single meter factor. It should first be determined from the Notice of Approval if the register has this feature, then whether it is in use. If it is, the service technician should be asked to record the existing values, and then enter a single factor for all flowrates. The fast and slow tests should be done as normal. The fast flow rate should be the maximum that the system is capable of, while the slow rate should be set as per Bulletin V3 section 4.3. On metering systems which will be used to measure more than one liquid, the test should be done using the liquid with the lowest viscosity. (ie., if the meter will be used to measure diesel and gasoline, the test should be done on gasoline).

4.2 Interpretation of results

For normal liquids, the difference between the results of the fast run and the slow run should not exceed 0.5% (2 × 0.25%). It is not necessary that both or either of the test results fall within + or − 0.25% of the true value. If the spread of the results exceeds 0.5%, testing should be terminated and normal procedures for non-compliant meters followed.

If the results do not differ by more than 0.5%, the multipoint calibration feature can be used. If multiple factors were determined during pretesting, they should be re-entered and all normal pertinent tests conducted. The meter must be adjusted as close as possible to zero error at all rates of flow, as per section 35.1 of the W&M regulations. The deviation between adjacent factors should be examined to ensure the requirements of SVM 1 part 10 are met.

For metering systems used on other than normal liquids, such as LPG or Lube oils, the above applies by analogy, with the exception that the spread should not exceed 2 times the applicable limit of error (LOE), rather than 0.5% and the results need not fall within the LOE.

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